

Hayato Saiki

PhD Student @ University of Tsukuba

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SportsXR | Human Computer Interaction | AR/VR | Computer Vision | Robotics

Education

University of Tsukuba

Ph.D. in Human Informatics

Ibaraki, Japan

Apr. 2024 – Now

- JSPS Research Fellowship for Young Scientists (DC2) (Advisor: Prof. [Kenji Suzuki](#))
- Research Fellow, Harvard University, John A. Paulson School of Engineering and Applied Sciences (Host: An Wang Prof. [Hanspeter Pfister](#))

University of Tsukuba

M.S. in Human Informatics

Ibaraki, Japan

Apr. 2022 – Mar. 2024

- Special Encouragement Student (Advisor: Prof. [Kenji Suzuki](#))

University of Tsukuba

B.S. in Engineering

Ibaraki, Japan

Apr. 2020 – Mar. 2022

National Institute of Technology, Suzuka College

Associate Degree in Electrical and Electronic Engineering

Mie, Japan

Apr. 2015 – Mar. 2020

Publications

International

- P1. **Hayato Saiki**, Chunggi Lee, Hikari Takahashi, Tica Lin, Kaori Tachibana, Hidetada Kishi, Yasuhiro Suzuki, Hanspeter Pfister, Kenji Suzuki, “**BRIDGE: Borderless Reconfiguration for Inclusive and Diverse Gameplay Experience via Embodiment Transformation**”, ACM Conference on Human Factors in Computing Systems, 2026 (CHI 2026, Overall acceptance rate: 25.3%) [[PDF](#)] [[Video](#)]
- P2. Chunggi Lee*, **Hayato Saiki***, Tica Lin, Eiji Ikeda, Kenji Suzuki, Chen Zhu-Tian, Hanspeter Pfister, “**ViSTAR: Virtual Skill Training with Augmented Reality with 3D Avatars and LLM Coaching Agent**”, ACM Conference on Human Factors in Computing Systems, 2026 (CHI 2026, Overall acceptance rate: 25.3%). [[PDF](#)] [[Video](#)]
- P3. **Hayato Saiki**, Masakazu Hirokawa, Modar Hassan, Kenji Suzuki, “**A Large-Scale Mixed Reality Stadium for Training Coordinated Tactical Plays in Basketball**”, Frontiers in Virtual Reality, 2025 [[PDF](#)]
- P4. **Hayato Saiki**, Kiyohiro Konno, Kosuke Naruse, Shimizu, Seiji Ono, Kenji Suzuki, “**Cognitive Load and Motor Adjustment Under Virtual Coaching Jump Shots in Basketball**”, IEEE International Conference on Systems, Man, and Cybernetics, 2023. [[PDF](#)]
- P5. Kazuma Takaishi, **Hayato Saiki**, Masakazu Hirokawa, Modar Hassan, Kenji Suzuki, “**A Sleeve Device Using Electrical Impedance for Coaching Jump Shots in Basketball**”, IEEE International Conference on Systems, Man, and Cybernetics, 2023. [[PDF](#)]

Preprint, Workshop, Poster, Domestic

- D1. **Designing Inclusive Future Augmented Realities**, The Association of Computing Machinery CHI Conference, 2024.
- D2. **Hayato Saiki**, Masakazu Hirokawa, Modar Hassan, Kenji Suzuki, “**A Large-Space Mixed Reality Stadium to Support Tactical Training in Basketball**”, The 1st Sports Informatics Research Presentation Symposium, Information Processing Society of Japan, 2024 (The Outstanding Presentation Award).
- D3. Yuki Tago, **Hayato Saiki**, Chihiro Ikuta, “**Particle Swarm Optimization Team Discovery**”, RISP International Workshop on Nonlinear Circuits, Communications and Signal Processing, 2023.
- D4. Kenta Tanaka, Mika Oki, **Hayato Saiki**, Kenji Suzuki, “**Interaction Analysis of Children with Neurodevelopmental Disorders in a Team-based Game Using Large-Scale Floor Projection**”, The 1st Sports Informatics Research Presentation Symposium, Information Processing Society of Japan, 2024.
- D5. **Hayato Saiki**, Masakazu Hirokawa, Modar Hassan, Kenji Suzuki, “**A Large-Space Mixed Reality Environment to Support the Improvement of Basketball Passing Skills**”, Proceedings of the 27th Annual Conference of the Virtual Reality Society of Japan, 2022.

Professional Experience

Harvard University (SEAS)

Boston, USA

Research Fellow (Host: An Wang Prof. [Hanspeter Pfister](#))

Feb. 2025 – Jan. 2026

- Conducted research in Human–Computer Interaction (HCI) focusing on **AI-driven XR systems for sports training**.
- Developed methods combining **Large Language Models (LLMs)** with **Random Forest classifiers** to enhance personalized feedback for athletes.
- Implemented computer vision pipelines to track players, the ball, and court geometry, enabling **3D reconstructions of basketball games and simulation under different embodiments**.
- Designed and evaluated a **reinforcement learning–based balancing system** across heterogeneous modalities (PC and AR environments) to ensure consistent training experiences.

National Agriculture and Food Research Organization (NARO)

Ibaraki, Japan

Part-time Researcher & Developer

Jun. 2022 – May 2023

- Developed an **AI-driven model for vehicle route optimization**, leveraging reinforcement learning techniques to improve autonomous path planning.
- Built and validated a **simulation environment for evaluating vehicle driving performance** across various scenarios.

Honors & Awards

Academic

- 2025 **JSPS Research Fellowship for Young Scientists**, JSPS Research Fellowship for Young Scientists (DC2)
| Highly competitive doctoral fellowship providing a monthly stipend and an annual research grant of ¥800,000
- 2025 **Japan Public-Private Partnership Student Study Abroad Program**, Ministry of Education, Culture, Sports, Science and Technology
| Highly competitive government-funded scholarship providing a preparation grant and monthly stipend
- 2025 **Support for Pioneering Research Initiated by the Next Generation (SPRING)**, Japan Science and Technology Agency (JST)
| Highly competitive doctoral fellowship providing a monthly stipend and an annual research grant of ¥500,000
- 2024 **The Outstanding Presentation Award**, The 1st Sports Informatics Research Symposium
- 2022 **Special Encouragement Student Program**, University of Tsukuba
| Full tuition exemption, monthly stipend

Athletics

- 2019 **Second Place (Captain)**, National Technical College Basketball Tournament
- 2019 **Champion (Captain)**, Tokai Technical College Basketball Tournament, advanced to National Competition
| 2015, 2018
- 2015 **Selected Player (U15)**, Mie Prefecture Elite Training Team
| One of only 15 players selected in Mie Prefecture; trained together with members of the Japan National Team
- 2014 **Champion & Outstanding Player Award**, Mie Prefecture Junior High School Basketball Tournament
- 2012 **Champion**, Mie Prefecture Qualifier, National Mini-Basketball Tournament, advanced to National Competition

Skills

Programming Languages:

C/C++, C#, Python, JavaScript, R, D3.js, Three.js, React, HTML, CSS

Design and Development Tools:

Unity, Figma, Adobe XD, Sketch, Miro, Tableau, Premiere

Demos & Videos

1. **ViSTAR video (CHI 2026)** [[Video](#)] <https://youtu.be/plykQOvpJEc>
2. **BRIDGE video (CHI 2026)** [[Video](#)] <https://youtu.be/6l75kgxwF-Q>
3. **HoopVista (NBA Analytics Tool)** [[Website](#)] <https://hoopvista.com/>